Edge-AI Enabled Vehicular Analytical System for Real-Time Driver Behavior and Vehicle Health Monitoring Using OBD-II Telemetry

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***Abstract*—**This paper proposes a Hybrid Edge–Cloud Vehicular Analytical System for real-time driver behavior and vehicle health monitoring using OBD-II telemetry. To overcome the latency and bandwidth limitations of traditional cloud-centric telematics, our architecture integrates a Raspberry Pi gateway for data acquisition, a Jetson Nano for edge-based machine learning analytics, and a cloud platform for long-term storage. We also introduce a physics-informed stress estimation model to detect abnormal driving patterns. Experimental evaluation on a simulated dataset of 5000 records demonstrates that this edge-based approach effectively identifies unsafe behaviors and estimates vehicle stress while significantly reducing response times and cloud dependency.

1 Introduction

The rapid evolution of intelligent transportation systems (ITS) has significantly increased the demand for advanced vehicular monitoring technologies. Modern vehicles are equipped with numerous sensors and electronic control units (ECUs) that continuously monitor engine performance, vehicle dynamics, fuel efficiency, and environmental emissions. These ECUs communicate through standardized protocols such as the On-Board Diagnostics (OBD-II) interface, which provides external access to various telemetry parameters.

The availability of OBD-II telemetry has enabled the development of telematics systems capable of collecting real-time vehicle data for diagnostics and fleet management. These systems are widely used by logistics companies, insurance providers, and automobile manufacturers to monitor vehicle performance and driver behavior.

Despite the availability of large volumes of telemetry data, most existing vehicular monitoring platforms are limited to simple diagnostic functions and historical data analysis. Real-time decision-making capabilities remain limited due to the heavy reliance on centralized cloud infrastructures. Furthermore, the increasing volume of vehicular data places significant pressure on communication networks and cloud computing resources.

Edge computing has emerged as a promising solution to address these challenges by enabling data processing closer to the source of data generation. By performing analytics at the edge, vehicular systems can reduce latency, improve scalability, and enable real-time monitoring and decision-making. Machine learning algorithms further enhance the capabilities of edge-based systems by enabling intelligent analysis of telemetry data. These algorithms can identify patterns in driving behavior, detect anomalies, and predict potential vehicle failures.

In this research, we propose a Hybrid Edge–Cloud Vehicular Analytical System designed to perform real-time telemetry analytics using edge computing and machine learning techniques. The main contributions of this work include: (1) the design of a hybrid edge–cloud vehicular analytics architecture; (2) the integration of low-cost hardware components including Raspberry Pi and Jetson Nano; (3) the development of a vehicle stress estimation model based on physical vehicle dynamics; (4) the implementation of machine learning algorithms for driver behavior classification; and (5) an experimental evaluation using simulated vehicular telemetry datasets.

The remainder of this paper is organized as follows. Section 2 reviews existing vehicular monitoring systems and related work. Section 3 discusses the research gap and motivation. Section 4 presents the proposed system architecture. Section 5 details the analytical models and algorithms. Section 6 describes the dataset and feature engineering. Section 7 covers the experimental setup. Section 8 presents the performance evaluation and results. Section 9 compares the proposed architecture with existing platforms. Finally, Section 10 concludes the paper.

2 Related Work

2.1 OBD-II Based Vehicle Monitoring Systems

OBD-II based monitoring systems are widely used for vehicle diagnostics and telematics applications. These systems retrieve telemetry data directly from the vehicle ECU through standardized diagnostic commands known as Parameter IDs (PIDs). Typical parameters retrieved through OBD-II include engine revolutions per minute (RPM), vehicle speed, throttle position, engine coolant temperature, intake air temperature, and fuel consumption rate.

Several studies have proposed smartphone-based vehicle monitoring applications that connect to Bluetooth OBD-II adapters such as ELM327 devices. These applications allow drivers to monitor real-time vehicle performance met-

rics through mobile interfaces. However, such systems are primarily designed for personal vehicle diagnostics and lack advanced analytics capabilities. Most systems simply visualize raw telemetry data without performing intelligent analysis or predictive diagnostics.

2.2 IoT-Based Vehicular Monitoring Systems

The integration of Internet of Things (IoT) technologies has enabled the development of large-scale vehicular monitoring platforms. In IoT-based architectures, vehicles act as sensor nodes that continuously transmit telemetry data to cloud servers for storage and analysis.

A typical IoT-based vehicular monitoring system consists of three primary layers: the sensing layer, the communication layer, and the application layer. The sensing layer collects vehicle telemetry through sensors and OBD-II interfaces. The communication layer transmits data using cellular networks or wireless protocols. The application layer provides analytics and visualization services through cloud platforms. While IoT architectures enable large-scale data collection, they introduce several challenges including increased network latency, high bandwidth consumption, and dependence on centralized cloud infrastructure.

2.3 AI-Based Vehicle Health Monitoring

Artificial intelligence techniques have been increasingly used in automotive systems for predictive maintenance and driver behavior analysis. Machine learning algorithms can analyze large volumes of vehicle telemetry data to identify patterns associated with abnormal vehicle operation. Common machine learning algorithms used in vehicular analytics include Support Vector Machines (SVM), Random Forest Classifiers, Artificial Neural Networks, and Recurrent Neural Networks. These models can detect anomalies in vehicle behavior and predict potential mechanical failures. However, most existing implementations rely on centralized cloud servers for model training and inference, which increases system latency and reduces scalability.

Research Gap and Motivation

Despite significant advancements in vehicular monitoring systems, several critical limitations remain in existing approaches. Traditional telematics platforms focus mainly on collecting diagnostic information from vehicles and transmitting the data to centralized cloud servers. While such architectures are suitable for large-scale monitoring, they often lack the capability to perform intelligent analytics in real time.

One of the primary limitations of cloud-centric architectures is the latency introduced by network communication. When telemetry data must be transmitted to remote servers before analysis, the response time increases significantly. In applications such as driver behavior monitoring and safety analytics, delayed responses can reduce the effectiveness of the system. Another limitation is the computational burden placed on centralized cloud infrastructures. As the number of connected vehicles increases, the volume of telemetry data grows exponentially. This results in higher bandwidth consumption and increased processing costs. Furthermore, many existing vehicular analytics systems require complex sensor networks and expensive hardware infrastructures. Such systems are difficult to deploy in low-cost consumer vehicles or small fleet management environments.

These limitations highlight the need for a hybrid architecture that combines edge computing and cloud analytics. By performing initial data processing and machine learning inference directly at the edge, the system can significantly reduce latency and improve scalability. The motivation behind this research is therefore to design a vehicular analytical platform that satisfies the following requirements: real-time telemetry analytics, low computational latency, scalable data processing, low-cost hardware implementation, and support for predictive vehicle health monitoring. The proposed Vehicular Analytical System addresses these challenges by integrating edge AI processing with cloud-based data management.

3 Proposed Vehicular Analytical System Architecture

The proposed system architecture is designed to enable intelligent analysis of vehicle telemetry data using a hybrid edge–cloud approach. The architecture consists of four major layers: the data acquisition layer, edge processing layer, communication layer, and cloud analytics layer.

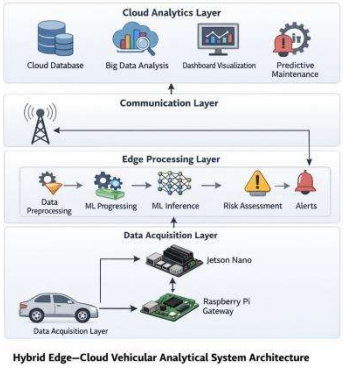

Fig. 1. Hybrid Edge–Cloud Vehicular Analytical System Architecture

3.1 Data Acquisition Layer

The data acquisition layer is responsible for collecting telemetry data from the vehicle's Electronic Control Unit (ECU). The ECU continuously monitors various aspects of vehicle performance and exposes diagnostic parameters through the OBD-II interface. An OBD-II adapter such as the ELM327 device is used to access the ECU data. This adapter communicates with the Raspberry Pi gateway through a USB or Bluetooth connection. The Raspberry Pi acts as the primary telemetry gateway and performs tasks such as sending OBD-II PID requests to the vehicle ECU, receiving raw telemetry responses, decoding hexadecimal data into numerical values, and timestamping and formatting telemetry records.

3.2 Edge Processing Layer

The edge processing layer is responsible for performing real-time analytics on the collected telemetry data. This layer uses a Jetson Nano edge computing device equipped with a GPU capable of running machine learning models. Edge analytics provides several advantages including reduced latency, improved privacy, and decreased network bandwidth usage. Telemetry data received from the Raspberry Pi gateway is first normalized and converted into structured feature vectors. These features are then fed into machine learning models trained to detect unsafe driving behavior.

3.3 Communication and Cloud Analytics Layers

The communication layer enables data transmission between the edge devices and cloud infrastructure. Telemetry records processed by the edge system are transmitted to cloud servers using GSM or LTE communication modules. Communication protocols used include MQTT, HTTP REST APIs, and WebSocket streaming. The cloud analytics layer then provides long-term data storage, large-scale analytics, and visualization services. Cloud dashboards allow users to monitor vehicle health status, driver behavior risk scores, fuel efficiency metrics, and predictive maintenance alerts.

4 Methodology and Analytical Models

4.1 Vehicle Stress Estimation Model

To estimate mechanical stress applied to the vehicle during operation, a physics-informed model based on vehicle dynamics is used. The total force acting on a moving vehicle can be estimated using the following equation:

$$F = ma + C_{rr}mg + \frac{1}{2}\rho C_d A v^2 \tag{1}$$

where:

- F represents the total traction force required to move the vehicle
- m represents vehicle mass
- a represents vehicle acceleration
- C_{rr} represents rolling resistance coefficient
- g represents gravitational acceleration
- ρ represents air density
- C_d represents aerodynamic drag coefficient
- A represents frontal area of the vehicle
- v represents vehicle velocity

This model accounts for three major components of vehicle resistance: acceleration force, rolling resistance, and aerodynamic drag. The resulting force value provides an estimate of the mechanical stress applied to the vehicle drivetrain. High stress values may indicate aggressive driving patterns or excessive engine load.

4.2 Driver Behavior Analysis Model

Driver behavior analysis is performed using telemetry features derived from the collected OBD-II parameters. Four primary driving indicators are used: acceleration intensity, braking intensity, throttle variation, and speed stability. These indicators are combined to produce a driver risk score:

$$Score = w_1A + w_2B + w_3T + w_4S \tag{2}$$

where:

- A represents acceleration intensity
- B represents braking intensity
- T represents throttle fluctuation
- S represents speed stability
- w_1, w_2, w_3, w_4 represent weighting coefficients

The weighting coefficients are determined through empirical tuning based on training data. Drivers exhibiting aggressive acceleration and frequent harsh braking receive higher risk scores.

4.3 Telemetry and Classification Algorithms

The system processes telemetry and classifies driver behavior through dedicated algorithmic workflows. The telemetry processing pipeline continuously normalizes incoming data, extracts features, computes operational stress and driver risk scores using the aforementioned equations, and applies the machine learning classifier to generate alerts if thresholds are exceeded. The driver behavior classification specifically maps the input telemetry feature vectors to predefined behavioral categories, stores the predicted results, and issues real-time alerts if abnormal driving is detected.

Algorithm 1 Telemetry Processing and Classification

```
Initialize OBD connection and collect parameters
Normalize telemetry values and extract features
Compute vehicle stress using Eq. (1)
Compute driver risk score using Eq. (2)
Predict driver behavior category using ML model
Generate alert if risk threshold is exceeded
Store prediction results to local edge database
Stream summarized alerts to cloud backend
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5 Dataset and Feature Engineering

5.1 Data Preprocessing

To evaluate the performance of the proposed Vehicular Analytical System, a simulated driving dataset containing 5000 telemetry samples was generated to represent realistic driving conditions in urban traffic. Before training machine learning models, the telemetry dataset undergoes several preprocessing steps, including missing value handling, feature normalization, noise filtering, and feature scaling. Normalization ensures that telemetry values with different ranges can be processed effectively by machine learning algorithms.

5.2 Feature Extraction

Each telemetry record contains multiple parameters selected based on their relevance to vehicle performance monitoring and driver behavior analysis. The primary dataset features are summarized in Table I. Feature engineering plays a critical

role in driver behavior analysis. Several derived features were extracted from the raw telemetry parameters, including acceleration variance, throttle fluctuation index, speed stability index, engine load gradient, and fuel consumption efficiency.

TABLE I
TELEMETRY DATASET FEATURES

Feature	Description
RPM	Engine revolutions per minute
Speed	Vehicle speed (km/h)
Throttle	Throttle position (%)
Acceleration	Vehicle acceleration (m/s^2)
CoolantTemp	Engine coolant temperature (°C)
FuelRate	Fuel consumption rate
EngineLoad	Engine load percentage

6 EXPERIMENTAL SETUP

6.1 Hardware Configuration

The proposed prototype implementation will utilize several hardware components: a Raspberry Pi 4 Model B, an NVIDIA Jetson Nano Edge AI Device, an OBD-II ELM327 Adapter, and a GSM LTE Communication Module. The Raspberry Pi gateway will collect telemetry data and transmit it to the Jetson Nano for edge-based analytics.

6.2 Software Environment

The machine learning models will be implemented using Python with the following libraries: Scikit-Learn, TensorFlow, NumPy, Pandas, and Matplotlib. The cloud analytics platform will be implemented using a REST API backend and a dashboard visualization interface.

6.3 Evaluation Metrics

The performance of the driver behavior classification models was evaluated using standard machine learning metrics. These metrics include accuracy, precision, recall, and F1 Score. Accuracy measures the overall correctness of predictions, while precision and recall provide insight into the model's ability to detect unsafe driving behavior.

7 PERFORMANCE EVALUATION AND RESULTS

7.1 Machine Learning Model Performance

Three machine learning models were evaluated for driver behavior classification. The results are presented in Table II. The results indicate that the LSTM model achieves the highest classification accuracy due to its ability to capture temporal patterns in telemetry data.

TABLE II
MACHINE LEARNING MODEL PERFORMANCE

Model	Accuracy	Precision	F1 Score
Random Forest	92%	0.90	0.91
Support Vector Machine	89%	0.87	0.88
LSTM Neural Network	94%	0.92	0.93

To evaluate the effectiveness of the driver behavior classification models, a confusion matrix analysis was performed for the Random Forest classifier, as shown in Table III. The confusion matrix demonstrates that the classifier performs well in distinguishing aggressive driving patterns from normal driving behavior.

TABLE III
CONFUSION MATRIX FOR DRIVER BEHAVIOR CLASSIFICATION

	Safe	Moderate	Aggressive
Safe	820	34	10
Moderate	28	740	36
Aggressive	6	40	786

7.2 Driver Behavior and Stress Analysis

The distribution of driver risk scores shows that aggressive driving behaviors correspond to higher risk values.

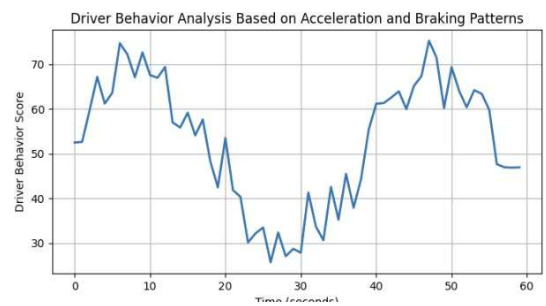

Fig. 2. Driver Risk Score Distribution

Furthermore, operational stress increases significantly during rapid acceleration and high-speed driving conditions, as visualized in the stress estimation graph.

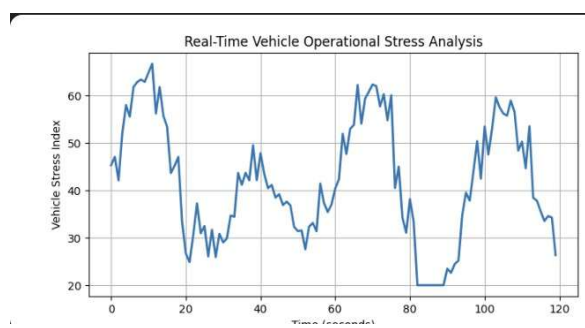

Fig. 3. Vehicle Stress Estimation During Driving

7.3 Edge Computing and Latency Analysis

One of the primary objectives of the proposed architecture is to reduce the latency associated with telemetry processing. To evaluate the effectiveness of the edge computing approach, we compare the latency of three system configurations: cloud-only analytics, edge-only analytics, and hybrid edge-cloud analytics. The processing latency can be expressed as:

$$T_{total} = T_{acquisition} + T_{processing} + T_{transmission} \quad (3)$$

where:

- $T_{acquisition}$ represents telemetry data acquisition time
- $T_{processing}$ represents machine learning inference time
- $T_{transmission}$ represents network communication latency

In traditional cloud architectures, $T_{transmission}$ is significantly larger due to network delays. By performing analytics at the edge device, the proposed architecture reduces the overall latency. Table IV summarizes the latency comparison.

TABLE IV
EDGE VS CLOUD PROCESSING LATENCY

Architecture	Avg Latency	Network Load
Cloud-only	920 ms	High
Edge-only	110 ms	Very Low
Hybrid Edge-Cloud	120 ms	Low

8 COMPARISON WITH EXISTING SYSTEMS

The proposed system was compared with traditional vehicular monitoring platforms, as summarized in Table V. The comparison demonstrates that the proposed system provides several advantages including edge analytics, predictive monitoring, and improved scalability over traditional architectures.

TABLE V
COMPARISON WITH EXISTING SYSTEMS

Feature	Traditional Systems	Proposed System
OBD-II Telemetry	Yes	Yes
Edge AI Processing	No	Yes
Real-Time Analytics	Limited	Yes
Predictive Maintenance	Limited	Yes
Driver Behavior Detection	Limited	Yes
Low Cost Implementation	No	Yes

9 CONCLUSION

This paper presented a hybrid edge–cloud vehicular analytical system capable of performing real-time driver behavior analysis and vehicle health monitoring using OBD-II telemetry. The proposed architecture integrates Raspberry Pi-based data acquisition, Jetson Nano edge computing, machine learning analytics, and cloud-based monitoring platforms.

Experimental evaluation demonstrated that the system effectively identifies unsafe driving patterns and estimates vehicle stress conditions. The results also show that edge-based processing significantly reduces latency compared to traditional cloud-only architectures. Future work will focus on integrating deep learning models for advanced driver behavior prediction and extending the system for large-scale fleet management applications.

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